

ME594 3D Robotics
HW8 Answer to selected questions

1)

$$\mathbf{M} = \begin{bmatrix} m_1 + m_2 & -\ell m_2 \cos \theta \\ -\ell m_2 \cos \theta & \ell^2 m_2 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} \ell m_2 \omega^2 \sin(\theta) \\ 0 \end{bmatrix}$$

$$\mathbf{G} = \begin{bmatrix} 0 \\ -g\ell m_2 \sin(\theta) \end{bmatrix}$$

2)

$$\mathbf{C} = \begin{bmatrix} 0.0148 \\ -0.0639 \\ 0.0248 \\ 0.0008 \\ 0.0022 \\ 0.0003 \end{bmatrix} \quad \mathbf{G} = \begin{bmatrix} -0.0000 \\ 1.8641 \\ -0.5640 \\ 0.0089 \\ -0.0273 \\ 0.0007 \end{bmatrix}$$

$$\mathbf{M} = \begin{bmatrix} 0.041718 & -0.000598 & -0.000503 & 0.003167 & 0.000044 & 0.002960 \\ -0.000598 & 0.077020 & 0.026277 & -0.000381 & 0.007060 & 0.000093 \\ -0.000503 & 0.026277 & 0.037292 & -0.000558 & 0.007376 & 0.000078 \\ 0.003167 & -0.000381 & -0.000558 & 0.003197 & 0.000021 & 0.003034 \\ 0.000044 & 0.007060 & 0.007376 & 0.000021 & 0.004322 & 0.000021 \\ 0.002960 & 0.000093 & 0.000078 & 0.003034 & 0.000021 & 0.003043 \end{bmatrix}$$